

Problem 1

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Problem 1 (a) Find the linearization, $L(x)$, of the function $f(x) = e^{2x}$ at $a = 0$.

Explanation. Recall that $L(x) = f(a) + f'(a)(x - a)$.
Hence $f'(x) = 2e^{2x} \implies f'(0) = 2e^0 = 2$ and $f(0) = e^0 = 1$.
Therefore $L(x) = 1 + 2 \cdot x$

(b) Using the linearization, $L(x)$, from the part (a), approximate e .

Explanation.

$$\begin{aligned} e &= e^{2 \cdot (1/2)} \\ &= f(1/2) \approx L(1/2) \\ &\approx 1 + 2 \cdot (1/2) \\ &\approx 2 \end{aligned}$$

(c) Is the estimation found in part (b) an overestimate or an underestimate?
EXPLAIN.

Explanation. $f''(x) = 4e^{2x}$ is positive on the interval $\left(0, \frac{1}{2}\right)$. That means the graph of f is concave up between $x = 0$ and $x = \frac{1}{2}$, so this estimate is an underestimate. (The graph of the linearization L lies below the graph of the function f .)